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Introduction

The growing demand of healthy-functional food, along with the increasing agriculture-associated environmental pressures are emphasizing the need for production of more nutritious food while applying sustainable practices¹. Utilization of edible wild plants might be a promising avenue to meet these needs². Wild species of the *Malvaceae* family, found in the Mediterranean area and named in the generic term "Khubeza", have been used as traditional food for many years. Our previous work showed that Khubeza leaves exhibited high nutritional value³. Here we aimed to study the nutritional value of pasta made of wheat flour fortified with Khubeza powder.

Methodology

Khubeza leaf powder- *Malva Nicaeensis* and *Lavatera Cretica* leaves were treated, freeze-dried and ground to fine powder that retains the nutritional and sensorial properties.

Pasta- Pasta was prepared by partially substituting of the wheat flour (control) with 10%, 15% and 20% of Khubeza powder.

Nutritional analysis- Cooked pasta were analyzed for the determination of: Total Dietary Fibers, Protein content, Total phenolics content, Ascorbic acid level and Mineral profile.



Figure 1: Khubeza plants at the ARO net house location.

Figure 2: Khubeza leaf powder.



Figure 3: Pasta based on partial replacement of wheat flour with various percentages of "Khubeza" powder.



Figure 4: Pasta tortellini dish, with 15% Khubeza powder.

Results

Enriching pasta with Khubeza powder resulted in a significant increase in protein (Fig. 5A) and dietary fibers (Fig. 5B) that reached 25% and 83%, respectively, in the treatment of 20% Khubeza powder as compared to the control pasta.

The levels of total polyphenols (Fig. 5C) and ascorbic acid (Fig. 5D) in the Khubeza pasta were increased in a dose-response manner up to three and four folds, respectively, in comparison to the levels in the control pasta.

M. nicaeensis higher than *L. cretica* in antioxidants, whereas slightly lower in protein (TDF- to be further investigated).

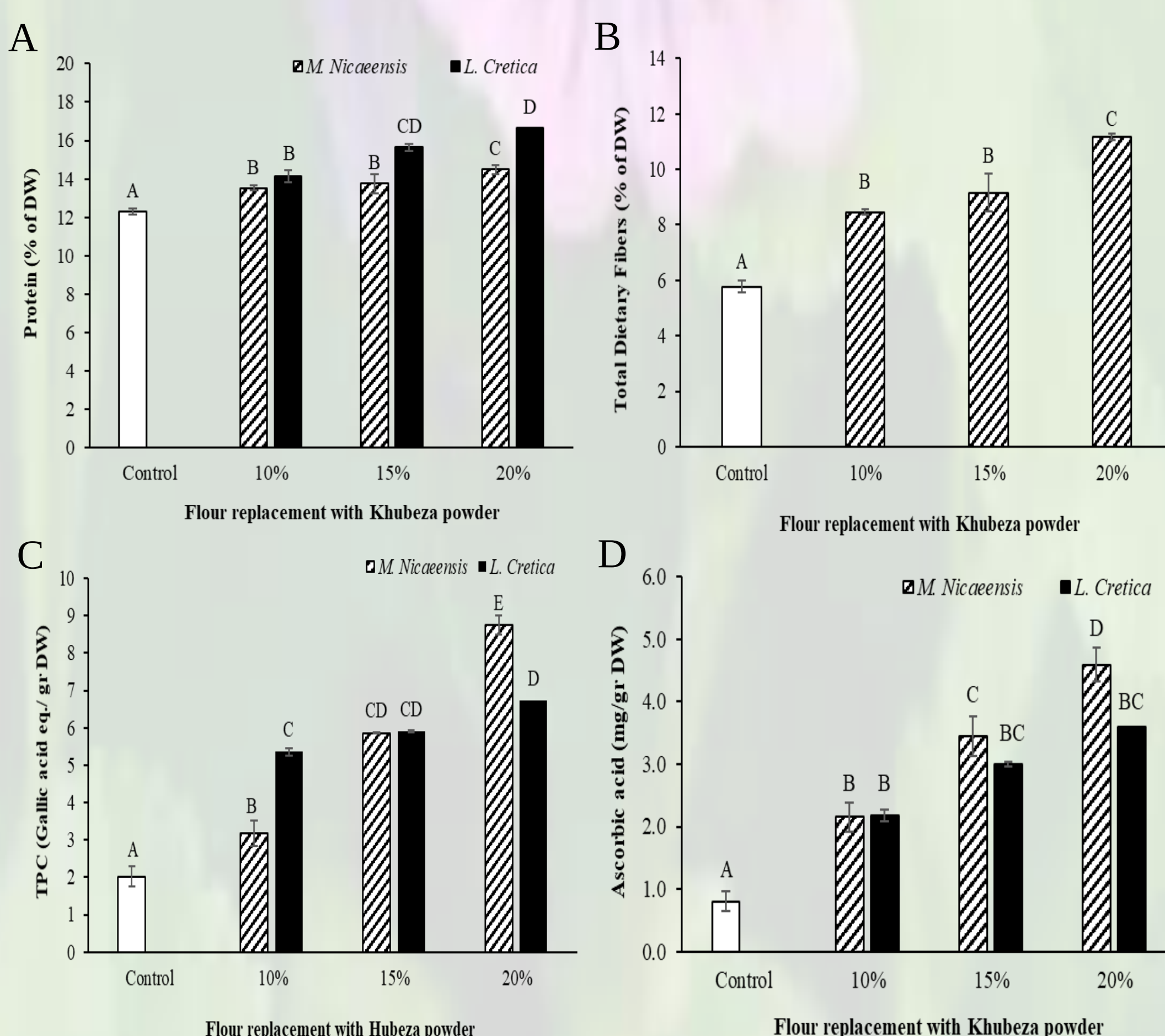
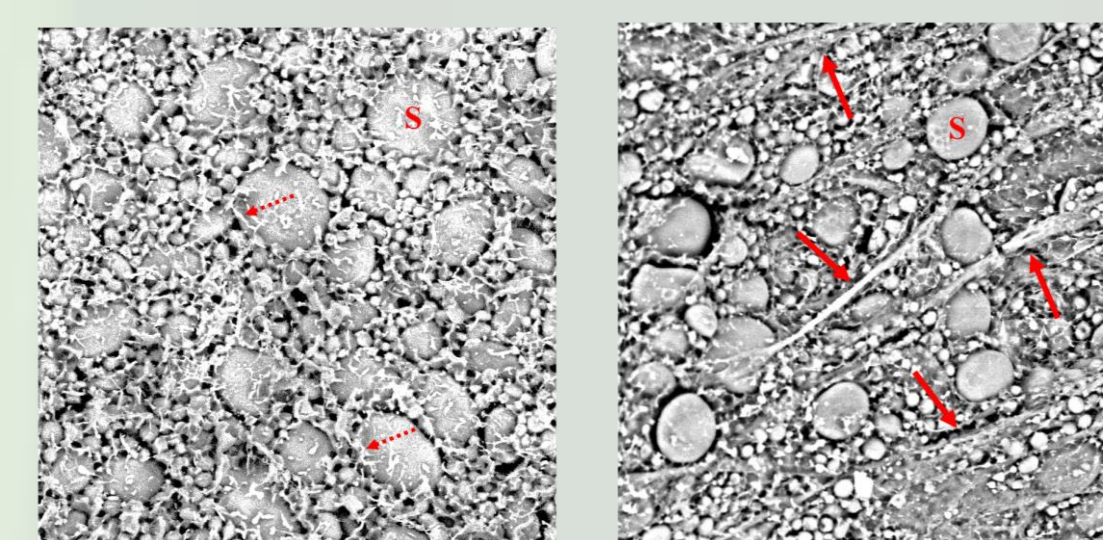


Figure 5: Nutritional parameters in pasta enriched with *M. Nicaeensis* and *L. Cretica* leaf powders - Protein (A), Total Dietary fibers (B), Total Phenolics Content (C) and Ascorbic acid (D). Data are means ± S.E.(n=3) and different letters indicate significant differences (P≤0.05).

The doughs SEM images demonstrate the effects of Khubeza powder on pasta structure: fewer starch globules (S), impaired glutenin net forming (dashed arrows) and appearance of fibers (continuous arrows) (Fig. 6).

Figure 6: SEM images of control dough (left) and dough enriched with 10% Khubeza powder (right). Magnification x1750



Substitution of wheat flour with Khubeza powder increased several minerals in the pasta, in a dose-response manner (Table 1).

Table 1- Main elements in pasta enriched with Khubeza powders. Concentrations expressed as (μg/gr DW), Data are means of duplicates:

Element	Control Pasta	Pasta enriched with various percentages of "Khubeza" powder					
		10%		15%		20%	
		<i>M. nicaeensis</i>	<i>L. cretica</i>	<i>M. nicaeensis</i>	<i>L. cretica</i>	<i>M. nicaeensis</i>	<i>L. cretica</i>
Ca	879.4	2904.6	2792.0	3960.8	377.0	4976.4	4557.9
Fe	14.5	23.4	22.5	29.4	24.9	33.1	28.2
K	828.6	1754.0	1624.1	1931.4	2002.1	1893.3	2634.1
Mg	418.0	570.6	682.0	671.6	760.1	760.2	823.1
P	1266.1	1501.4	1717.4	1745.1	1778.4	1865.0	1914.4

Conclusions

These findings, measured in the cooked pasta to imitate the actual consumption manner, emphasizes the nutritional relevance and beneficial contribution inherent in the use of Khubeza for nutritional fortification of bakery and pasta products.

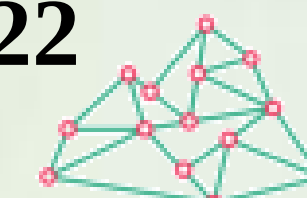
Due to its nutritional and agricultural advantages, domestication of Khubeza may contribute to food security and serve as a novel food crop.

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Conference

Innovation in Mediterranean Traditional Foods: novel products and processes (IMTF2022), Bragança, Portugal, October 13-14, 2022



montanhas de investigação

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